

Sungbok Shin 

(Incoming) Assistant Professor at **Sogang University**

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Research Interests

I develop methods and systems that augment our everyday activities using Artificial Intelligence (AI).
As a first step, I explored methods and tools that use AI to augment visualization design process.
I am also interested in topics on Human-Centered AI (HCAI), Data Visualization (InfoVis, VA), HCI, Biomedical Informatics, and Situated Analytics (AR).

Education

University of Maryland, College ParkCollege Park, MD, United States

Ph.D. in Computer Science (Advised by **Dr. Niklas Elmqvist**)8.2019 ~ 8.2024

Dissertation: **Simulation, Representation, and Automation: Human-Centered AI for Augmenting Visualization Design**

Committee: Niklas Elmqvist (Chair), Philip Resnik, Zhicheng Liu, Christopher Metzler, and Sanghyun Hong.

Korea UniversitySeoul, South Korea

M.Eng. in Computer Science and Engineering (Advised by **Dr. Jaegul Choo**)3.2017 ~ 2.2019

Thesis: Spatio-Temporally Exclusive Topic Discovery for Anomalous Event Detection

B.S. in Computer Science and Engineering, Mathematics (Advised by **Dr. Sungwoo Chung**)3.2009 ~ 2.2017

Professional Career

Sogang UniversitySeoul, South Korea

Assistant Professor2026.3 (Incoming)

Dept. of Computer Science and Engineering,
College of Computing

Aarhus UniversityAarhus, Denmark

Postdoctoral Fellow (PI: **Dr. Niklas Elmqvist**)10.2025 ~ (now)

Seoul National UniversitySeoul, South Korea

Visiting Postdoctoral Scholar (PI: **Dr. Jinwook Seo**)9.2025 ~ 9.2025

Inria (Aviz, Saclay Center)Paris, France

Postdoctoral Fellow (PI: **Dr. Jean-Daniel Fekete**)9.2024 ~ 8.2025

University of Maryland, College ParkCollege Park, MD, United States

Graduate Research/Teaching Assistant8.2019 ~ 8.2024

Research Assistant – (DataWorld – Situated Analytics)F 2019, S/F 2020, S 2021, F 2023

Teaching Assistant – (CMSC250 – Discrete Structures, CMSC320 – Introduction to Data Science)F 2021, S/F 2022, S 2023, S 2024

Korea UniversitySeoul, South Korea

Under-/Graduate Research Assistant10.2016 ~ 7.2019

Full Paper/Journal Publications (Peer-reviewed)

Enriching Semantic Profiles into Knowledge Graph for Recommender Systems Using Large Language Models
Seokho Ahn, **Sungbok Shin***, Youngduk Seo* (*:Co-correspondence)
ACM Conference on Knowledge Discovery and Data Mining (KDD’2026), 2026, Jeju, South Korea [BK IF=4]

Dataset-Adaptive Dimensionality Reduction
Hyeon Jeon, Jeongin Park, Soohyun Lee, Dae Hyun Kim, **Sungbok Shin**, Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (VIS2025’), Vienna, Austria (In Press) [IF=6.5, Top 6% in CS/SWE]

Visualizationary: Automating Design Feedback for Visualization Designers Using Large Language Models
Sungbok Shin, Sanghyun Hong, Niklas Elmqvist
IEEE Transactions on Visualization and Computer Graphics (TVCG), 2025, (In Press) [IF=6.5, Top 6% in CS/SWE]

Drillboards: Adaptive Visualization Dashboards for Dynamic Personalization of Visualization Experiences

Sungbok Shin*, Inyoup Na*, Niklas Elmqvist (*:Equal contributions)

IEEE Transactions on Visualization and Computer Graphics (TVCG), 2025, (In Press) [IF=6.5, Top 6% in CS/SWE]

Conversation Progress Guide: UI System for Enhancing Self-Efficacy in Conversational AI

Daeun Jung, **Sungbok Shin**, Jongwook Jeong

ACM Conference on Human Factors in Computing Systems (CHI2025'), 2025, Yokohama, Japan [BK IF=4]

Real-time Calibration Model for Low-cost Sensor in Fine-grained Time Series

Seokho Ahn, Hyungjin Kim, **Sungbok Shin**, Young-Duk Seo

AAAI Conference on Artificial Intelligence (AAAI), 2025, Philadelphia, PA, United States [BK IF=4]

The Reality of the Situation: A Survey of Situated Analytics

Sungbok Shin, Andrea Batch, Peter W. S. Butcher, Panagiotis D. Ritsos, Niklas Elmqvist

IEEE Transactions on Visualization and Computer Graphics (TVCG), 30(8), 5147-5164, 2024 (Presented at VIS2023') [IF=6.5, Top 6% in CS/SWE]

Evaluating View Management for Situated Visualization in Web-based Handheld AR

Andrea Batch, **Sungbok Shin**, Julia Liu, Peter W. S. Butcher, Panagiotis D. Ritsos, Niklas Elmqvist

Computer Graphics Forum (EuroVis2023'), 2023, Leipzig, Germany [IF=2.7]

Perceptual Pat: A Virtual Human Visual System for Iterative Visualization Design

Sungbok Shin, Sanghyun Hong, Niklas Elmqvist

ACM Conference on Human Factors in Computing Systems (CHI2023'), 811:1~17, 2023, Hamburg, Germany [BK IF=4]

A Scanner Deeply: Predicting Gaze Heatmaps on Visualizations Using Crowdsourced Eye Movement Data

Sungbok Shin, Sunghyo Chung, Sanghyun Hong, Niklas Elmqvist

IEEE Transactions on Visualization and Computer Graphics (VIS2022'), 29(1), 396-406, 2023, Oklahoma City, OK, United States [IF=5.2]

TopicOnTiles: Tile-Based Spatio-Temporal Event Analytics via Exclusive Topic Modeling on Social Media

Minsuk Choi, **Sungbok Shin**, Jinho Choi, Scott Langevin, ..., and Jaegul Choo

ACM Conference on Human Factors in Computing Systems (CHI2018'), 583:1--583:11, 2018, Montreal, QC, Canada [BK IF=4]

Localized User-Driven Topic Discovery via Boosted Ensemble of Nonnegative Matrix Factorization

Sangho Suh, **Sungbok Shin**, Joonsuk Lee, Chandan K. Reddy, and Jaegul Choo

Knowledge and Information Systems (KAIS), 503-531, 2018 [IF=2.4]

STExNMF: Spatio-Temporally Exclusive Topic Discovery for Anomalous Event Detection

Sungbok Shin, Minsuk Choi, Jinho Choi, Scott Langevin, ..., and Jaegul Choo

IEEE International Conference on Data Mining (ICDM2017'), 435-444, 2017, New Orleans, LA, United States [BK IF=4]

Publications form Workshop/Abstract/Short Paper Proceedings

Understanding Bias in Perceiving Dimensionality Reduction Projections

Seoyoung Doh, Hyeon Jeon, **Sungbok Shin**, Ghulam Jilani Quadri, Nam Wook Kim, Jinwook Seo

IEEE VIS2025' Workshop on Visualization for Communication (VisComm), 2025, Vienna, Austria

Changing the Paradigm from Dynamic Queries to LLM-generated SQL Queries with Human Intervention

Ambre Assor, Hyeon Jeon, **Sungbok Shin**, Jean-Daniel Fekete

IEEE VIS2025' 16th Workshop on Visual Analytics in Healthcare (VAHC), 2025, Vienna, Austria

Libraries' Approaches to the Security of Public Computers

Samual Dooley, Michael Rosenberg, Eric Sloate, **Sungbok Shin**, Michelle Mazurek

USENIX SOUPS Workshop on Inclusive Privacy and Security (WIPS), 2020

VATUN: Visual Analytics for Testing and Understanding Convolutional Neural Networks

Chunbok Park, Soyoung Yang, Inyoup Na, Sunghyo Chung, **Sungbok Shin**, Bum Chul Kwon, Deokgun Park, Jaegul Choo

EuroVis2021' Short Paper, 2021, Zurich, Switzerland

SANVis: Visual Analytics for Understanding Self-Attention Networks

Chunbok Park, Inyoup Na, Yongjang Jo, **Sungbok Shin**, ..., and Jaegul Choo
IEEE VIS2019' Short Paper, 2019, Vancouver, BC, Canada

Presentations

- 1. IEEE ICDM2017' Conference, Nov 2017, New Orleans, LA, United States, Author
- 2. IEEE VIS2022' Conference, Oct 2022, Oklahoma City, OK, United States, Author
- 3. CS @ Inha University, Jan 2023, Incheon, South Korea, Small Talk
- 4. ACM CHI2023' Conference, Apr 2023, Hamburg, Germany, Author
- 5. HCIL Symposium @ University of Maryland, May 2023, College Park, MD, United States, Small Talk
- 6. HAIV Lab @ UNIST, Aug 2023, Ulsan, South Korea, Talk
- 7. IEEE VIS2023' Conference, Oct 2023, Melbourne, Australia, Author
- 8. Data Science @ Inha University, Nov 2023, Online, Online Seminar
- 9. Interactive Data Computing Lab @ SKKU, Jan 2024, Suwon, South Korea, Small Talk
- 10. HVCL Lab @ Korea University, Jan 2024, Seoul, South Korea, Lab Talk
- 11. VISUS Group @ University of Stuttgart, Jan 2024, Stuttgart, Germany, Weekly Colloquium
- 12. Sejong University, Jul 2024, Seoul, South Korea, Small Talk
- 13. HAIV Lab @ UNIST, Aug 2024, Ulsan, South Korea, Workshop on Visualization
- 14. Hanyang University, Jul 2024, Seoul, South Korea, Lab Talk
- 15. Team Aviz @ Inria, Oct 2024, Saclay, France, Weekly Colloquium

Awards and Fellowships

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| 1. <i>Dean's Fellowship</i> for outstanding academic achievement (\$2,500/yr, 2 years), 2019 | University of Maryland |
| 2. <i>Graduate Summer Fellowship</i> (\$5,000 for Summer 23'), 2023 | University of Maryland |

Reviewing

External Reviewer	ACM CHI 2024' 2025', 2026'
	IEEE VIS 2024' 2025'
	IEEE TVCG 2024' 2025'

Mentored Students

- 1. Seokho Ahn, Ph.D. Student @ Inha University (2024 ~ 2025)
- 2. Joonhyeok Kwon, Ph.D. Student @ POSTECH (2024 ~ Now)

Services

- 1. *Student Volunteer*, IEEE VIS2022' Conference, Oklahoma City, OK, United States, Oct. 2022
- 2. *Student Volunteer*, ACM CHI2023' Conference, Hamburg, Germany, Apr. 2023
- 3. *Student Volunteer*, IEEE VIS2023' Conference, Melbourne, Australia, Oct. 2023

Skills

Programming languages	Python, Matlab, Javascript
Languages	Native in Korean
	Fluent in oral and written in English
	Conversant in French

References

Dr. Niklas Elmqvist	Dept. of Computer Science, Aarhus University, Denmark (elm@cs.au.dk)
Dr. Sanghyun Hong	Dept. of EECS, Oregon State University, United States (sanghyun.hong@oregonstate.edu)
Dr. Jaegul Choo	Graduate School of AI, KAIST, South Korea (jchoo@kaist.ac.kr)

Dr. **Youngduk Seo**

Dept. of ECE, Inha University, Incheon, South Korea (mysid88@inha.ac.kr)